FRAGO-URBAN-OS

Programmable City

可编程城市

Jing-Zhang AI Belt urban governance protocol design

The Jing-Zhang AI innovation belt is not a city that "uses AI to assist planning". It is a city whose governance is designed as an Agent OS — participation, approval, piloting, evaluation and knowledge capture are all turned into protocols, driven by events and fully auditable. AI and citizens collaborate on one shared protocol, and humans keep the final decision.

Overall design area (submitted boundary)	11.413 sq km
Key detailed-design areas	3 areas · 369.3 ha total
Land use	58 polygons · gap-free, overlap-free, fully tiling the boundary
Boundary status	provisional · no official redline yet; the whole package must be recomputed once released

Concept: the city as an operating system that keeps running

frago's hook / def / recipe / human final judgement, mapped layer by layer onto urban governance

- **Sense layer (hook) urban sensing and response**
  Urban events — footfall, energy, AI service calls, public feedback — trigger the matching governance mechanism automatically, instead of stacking manual approvals. In space this maps to sensing-facility R&D land, road corridors and scenario pilot units.
- **Knowledge layer (def) the city knowledge base**
  Planning material, standards and data are captured as queryable, traceable knowledge domains. AI and citizens draw from the same source, ending information asymmetry. In space this maps to research and education land and open-data access points.
- **Process layer (recipe) public affairs workflows**
  Participation, piloting, evaluation and approval become transparent, auditable standard processes that anyone can copy, review and improve. In space this maps to the service interface belt and the protocol commercialisation window.
- **Human adjudication layer humans keep the final judgement**
  Every mechanism has a human adjudication gate: machines advise and execute but never replace decisions. In space this maps to housing and community service land, protocol living rooms and protocol reserve units.

A single narrative language

Treat the city as an operating system that keeps running: planning is not a one-off blueprint but a set of iterable protocols plus a continuously running governance loop.

In space that loop is —
one protocol bus: the Jing-Zhang heritage spine running the full corridor, about 29.6 km of unbroken slow mobility;
three protocol loops: closed slow-mobility rings inside the key areas, with the four layers arranged along them;
eight protocol nodes: public interfaces shared by release, review, display and deliberation;
one transverse section: the four protocol layers stacked from west to east.

Discipline

No conclusions on floor area ratio, building height, development intensity, retain-renovate-demolish, road alignment, rail alignment, bridges, tunnels or underground feasibility. Every AI scenario is labelled pending human review, conceptual and not approved for operation. Every cited figure carries a source marker.

Naming and visual identity

Chinese name	可编程城市
English name	Programmable City
Brand subtitle	Jing-Zhang AI Belt urban governance protocol
Visual direction	Protocol / loop / layered abstraction; cool technical teal plus the industrial rail-brown of the Jing-Zhang railway
Logo direction	A rail track growing vertically into a layered urban section (track = the century-old Jing-Zhang works, layers = protocol layers)

Three scope levels and the geographic facts

All inherited from provisional boundaries inferred from the official announcement; areas recomputed in EPSG:4548 and checked line by line against announced figures

Scope	Announced area	Recomputed here	Deviation	Basis and role
Coordinated research area	4,360.00 ha	**4,360.92 ha**	+0.02%	North to N 5th Ring Rd, east to Jingzang Expressway, south to Xizhimenwai St, west to Wanquanhe Rd (written extents from the announcement)
Overall design area (submitted boundary)	1,140.00 ha	**1,141.28 ha**	+0.11%	1 - 2 km around the Jing-Zhang heritage park; north to N 5th Ring Rd, east to Xueyuan Rd / Xitucheng Rd, south to Xizhimenwai St, west to Dazhongsi E Rd / Heqing Rd
Key areas, total (three)	368.40 ha	**369.29 ha**	+0.24%	North to south: Zhongzhiyuan AI Innovation Acceleration Area, Beijing AI Origin Community, Dazhongsi AI Industry Cluster
└ Zhongzhiyuan AI Innovation Acceleration Area	192.10 ha	**192.92 ha**	+0.43%	Protocol R&D and compute base; carries the sense (hook) and knowledge (def) layers
└ Beijing AI Origin Community	104.30 ha	**104.32 ha**	+0.02%	Protocol testbed with real residents; carries the human adjudication and process (recipe) layers
└ Dazhongsi AI Industry Cluster	72.00 ha	**72.05 ha**	+0.06%	Protocol commercialisation and showcase window; carries the process (recipe) and knowledge (def) layers

Boundary status and replacement conditions

No downloadable official polygon, CAD or GIS redline with a verifiable coordinate system is publicly available. The three scope levels and the three key areas are inherited verbatim from brief/site-package/geometry/provisional_boundaries.geojson, marked geometry_role=provisional_constraint, official_boundary=false, boundary_precision=provisional_rough. The rectangular key-area edges must not be read as parcel redlines, road redlines or tenure boundaries. Once prequalification documents, a formal brief or other cleared official data is obtained, all three scope levels and all three key areas must be replaced together and every layer, drawing, HTML page and metric regenerated — replacing a single boundary file is not enough.

Programmable City

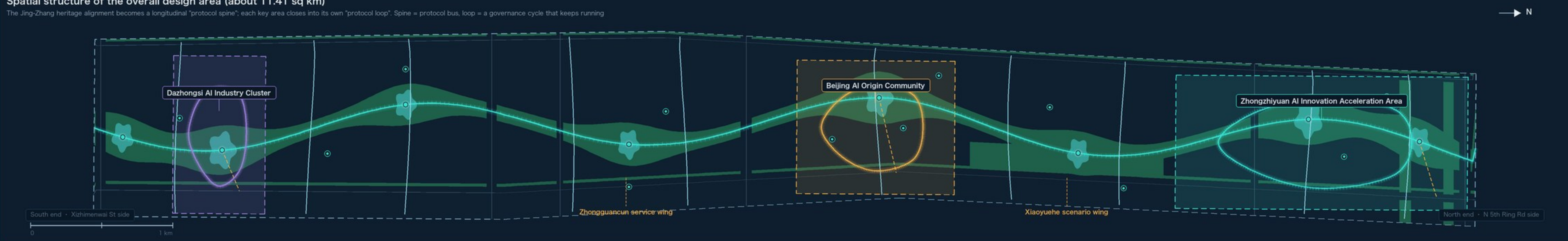
Jing-Zhang AI Belt urban governance protocol design | Overall spatial structure: layers (four-layer protocol section) + loops (one spine · three loops · eight protocol nodes)

FIG 01 / SITE OVERVIEW

PROVISIONAL · not an official redline

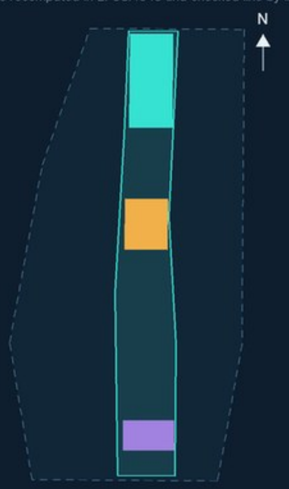
Spatial structure of the overall design area (about 11.41 sq km)

The Jing-Zhang heritage alignment becomes a longitudinal "protocol spine"; each key area closes into its own "protocol loop". Spine = protocol bus, loop = a governance cycle that keeps running



Three scope levels and their nesting

All areas recomputed in EPSG:4548 and checked line by line against announced figures



Coordinated research area
Announced about 4,360.0 ha | recomputed 4,360.9 ha | deviation +0.02%

Overall design area (submitted boundary)
Announced about 1,140.0 ha | recomputed 1,141.3 ha | deviation +0.11%

Key areas, total (three)
Announced about 368.4 ha | recomputed 369.3 ha | deviation +0.24%

Three key detailed-design areas

- **Zhongzhiyuan AI Innovation Acceleration Area 192.9 ha**
  Protocol R&D and compute base | sense (hook) + knowledge (def)
- **Beijing AI Origin Community 104.3 ha**
  Protocol testbed (with real residents) | human adjudication + process (recipe)
- **Dazhongsi AI Industry Cluster 72.0 ha**
  Protocol commercialisation and showcase | process (recipe) + knowledge (def)

Provisional boundary note: no official redline has been released. All three scope levels and the three key areas are inherited from the repository file provisional_boundaries.geojson with geometry_role=provisional_constraint and official_boundary=false. They must not be used for approval or precise-area claims.

Four-layer city OS architecture (layering)

frago's hook / def / recipe / human final judgement, mapped layer by layer onto urban governance

Sense layer `[hook]`
Urban events trigger governance automatically
Footfall / energy / AI service calls / public feedback

Knowledge layer `[def]`
Planning material and standards captured as queryable knowledge domains
AI and citizens draw on the same source, ending information asymmetry

Process layer `[recipe]`
Standardised participation / piloting / evaluation / approval
Anyone can copy, review and improve them

Human adjudication `[human]`
Every mechanism has a human adjudication gate
Machines advise and execute; they do not replace decisions

LOOP

Planning is not a one-off blueprint but a set of iterable protocols plus a continuously running governance loop: decisions are written back as new events and knowledge, which trigger the next round automatically. Spatially, that loop is the spine plus the three protocol loops.

Roles of the three areas and two wings under the protocol

Not conventional functional zoning — the same protocol playing different roles at different stages

Zhongzhiyuan AI Innovation Acceleration Area
Protocol R&D and compute base
Layers carried: sense (hook) + knowledge (def)
192.9 ha (provisional)

Beijing AI Origin Community
Protocol testbed (with real residents)
Layers carried: human adjudication + process (recipe)
104.3 ha (provisional)

Dazhongsi AI Industry Cluster
Protocol commercialisation and showcase window
Layers carried: process (recipe) + knowledge (def)
72.0 ha (provisional)

Zhongguancun service wing
The protocol's service layer: government services packaged as standard APIs

Xiaoyuehe scenario wing
A live testbed for scenarios: one public experience route threading them all

Corridor section: four protocol layers on one cut (layering)

Conceptual section across the corridor from west to east; widths are indicative and imply no cross-section, setback or engineering conclusion

Buffer green	Existing urban fabric (west)	Protocol spine	Existing urban fabric (east)	Service and scenario belt	Buffer green + road corridor
Boundary buffer	Knowledge layer (def): universities · institutes · education	Loop layer: heritage green spine + slow-mobility bus + protocol living room	Human adjudication layer: residents · community services · reserve	Process layer (recipe): service interfaces · scenario pilots	Sense layer (hook): traffic · stations · event capture

West · Dazhongsi E Rd - Heqing Rd side
East · Xueyuan Rd - Xitucheng Rd side

The concept in one sentence: the Jing-Zhang AI innovation belt is not a city that "uses AI to assist planning" — it is a city whose governance is designed as an Agent OS. Participation, approval, piloting, evaluation and knowledge capture all become protocols: event-driven and auditable. AI and citizens work on one shared protocol, and humans keep the final decision.

All spatial proposals are conceptual suggestions / reference schemes / material for professional teams to develop further. No conclusions are given on floor area ratio, building height, development intensity, retain-renovate-demolish, road alignment, rail alignment, bridges, tunnels or underground feasibility. Every AI scenario is labelled "pending human review", "conceptual illustration", "not approved for operation".

Phasing as protocol version management

Protocol release phase 474 ha · 42%	Pilot phase 406 ha · 36%	Extension phase 261 ha · 23%
Publish protocol text and knowledge domains; three key-area cores + first spine segment	Wings and spine-flanking blocks join; human review gates tested in practice	Existing fabric joins progressively; reserve units decided jointly by citizens and professional teams

Core metrics (computed from the geometry)

Displayed values match metrics.json exactly

Submitted boundary area	Land-use coverage
11.41 sq km	100.0 %

Green ratio (self-computed)	Public space ratio
19.3 %	6.2 %

Legend

- Protocol spine / slow-mobility bus
- Protocol node plazas (8)
- Rail station connection (indicative)
- Park green and buffer green
- Jing-Zhang railway heritage alignment (indicative)
- Submitted boundary (provisional)
- Coordinated research area (provisional)
- Key detailed-design areas

Sources: Haidian Branch, Beijing Municipal Commission of Planning and Natural Resources announcement (DATA-SRC-OFFICIAL-ANNOUNCEMENT20260509) + repository provisional boundaries provisional_boundaries.geojson (DATA-SRC-PROVISIONAL-BOUNDARIES20260605); spatial features and metrics recomputed from this proposal's geometry/*.geojson in EPSG:4548.
Status: the three scope levels and three key areas are provisional rough boundaries (official_boundary=false). Once an official redline is released, all layers, drawings, metrics and HTML must be recomputed as one package.
Metric basis: coordinates exchanged in EPSG:4326, areas recomputed in EPSG:4548. Every value on this sheet comes from metrics.json and can be verified with scripts/spatial_review.py.

Land-use structure of three areas and two wings

A gap-free, overlap-free, fully tiled partition cut piece by piece from the submitted boundary, with adjacent polygons sharing vertices

Land Use Structure

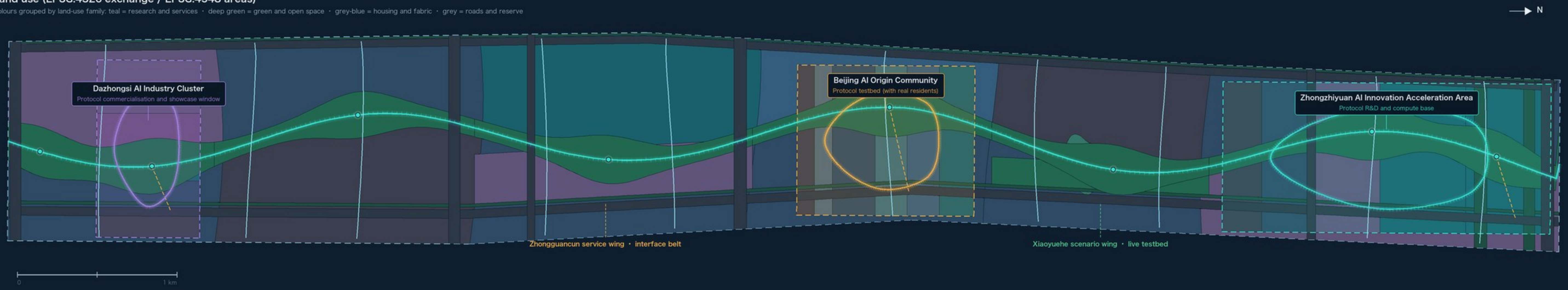
Land use is cut piece by piece from the submitted boundary: gap-free, overlap-free, fully tiled; adjacent polygons share vertices, union(land_use) = site_boundary

FIG 02 / LAND USE

PROVISIONAL · not an official redline

Land use (EPSG:4326 exchange / EPSG:4548 areas)

Colours grouped by land-use family: teal = research and services · deep green = green and open space · grey-blue = housing and fabric · grey = roads and reserve



Land-use composition

Areas computed from geometry/land_use.geojson in EPSG:4548; shares use the submitted boundary as denominator

Land use	Area	Share
Urban housing (Code 0701)	217.33 ha	19.04%
Reserved land (Code 16)	186.31 ha	16.32%
Park green (Code 1401)	179.59 ha	15.74%
Research (Code 0802)	159.15 ha	13.95%
Urban road land (Code 1207)	144.48 ha	12.66%
Commercial and business services (Code 05)	128.89 ha	11.29%
Buffer green (Code 1402)	41.14 ha	3.60%
Education (Code 0804)	33.65 ha	2.95%
Plaza (Code 1403)	20.33 ha	1.78%
Culture (Code 0803)	11.64 ha	1.02%
Urban community service facilities (Code 0702)	10.15 ha	0.89%
Health facilities (Code 0806)	8.00 ha	0.70%
Sports (Code 0805)	0.63 ha	0.06%
Total	1,141.28 ha	100.00%

Reserved land (16) accounts for 16.3%. This is a deliberate "protocol reserve": use and intensity are not fixed in this round but left for citizens and professional teams to decide together under the protocol — the direct land-use counterpart of " humans keep the final decision".

Structural logic: why this is not conventional colour-block zoning

One protocol: a bus along the length, a section across the width, a loop within each key area

Along: one protocol bus

The Jing-Zhang heritage alignment is abstracted into a continuous green spine with about 29.6 km of unbroken slow mobility, threading the three areas and two wings onto one bus so that an event on any segment travels to the rest.

Across: four-layer protocol section

From west to east: buffer green, knowledge-layer fabric, protocol spine, human-adjudication-layer fabric, process service belt, sense-layer road corridor. The layering is not formal decoration — it places different governance functions where they can see one another.

Local: three closed loops

Each key area closes into a slow-mobility loop with the four layers arranged along it, so that "propose — pilot — evaluate — decide — capture" can run repeatedly at walking scale.

Reserve: clauses not yet decided

186 ha of reserved land sits across the wings and the fabric, standing for clauses of the governance protocol not yet voted on — so a one-off blueprint does not lock in the future.

Land-use legend and topology self-check

Land-use codes (project subset of the national Land and Sea Use Classification Guideline)

0701 Urban housing	0802 Research	1402 Buffer green	0803 Culture	0805 Sports
16 Reserved	1207 Urban roads	0804 Education	0702 Community service facilities	
1401 Park green	05 Commercial services	1403 Plaza	0806 Health	

gap between union(land_use) and the boundary	0.47 sqm	union(land_use) outside the boundary	2.96 sqm
Largest pairwise overlap between land-use polygons	0.02 sqm	Threshold (set by this proposal, stricter than the repository's 1141 sqm)	11.5 sqm

Sources: Haidian Branch, Beijing Municipal Commission of Planning and Natural Resources announcement (DATA-SRC-OFFICIAL-ANNOUNCEMENT20260509) + repository provisional boundaries provisional_boundaries.geojson (DATA-SRC-PROVISIONAL-BOUNDARIES20260605); spatial features and metrics recomputed from this proposal's geometry/*.geojson in EPSG:4548.
Land-use designations are conceptual suggestions. They do not constitute land-use approval, tenure judgement, change of land-use classification, or retain/renovate/demolish conclusions. Existing-fabric features carry geometry_role=existing_condition and confidence=low.
Road corridors are located roughly from public road names, used only to express structural relationships. They imply no road redline, alignment or cross-section conclusion.

Key Areas

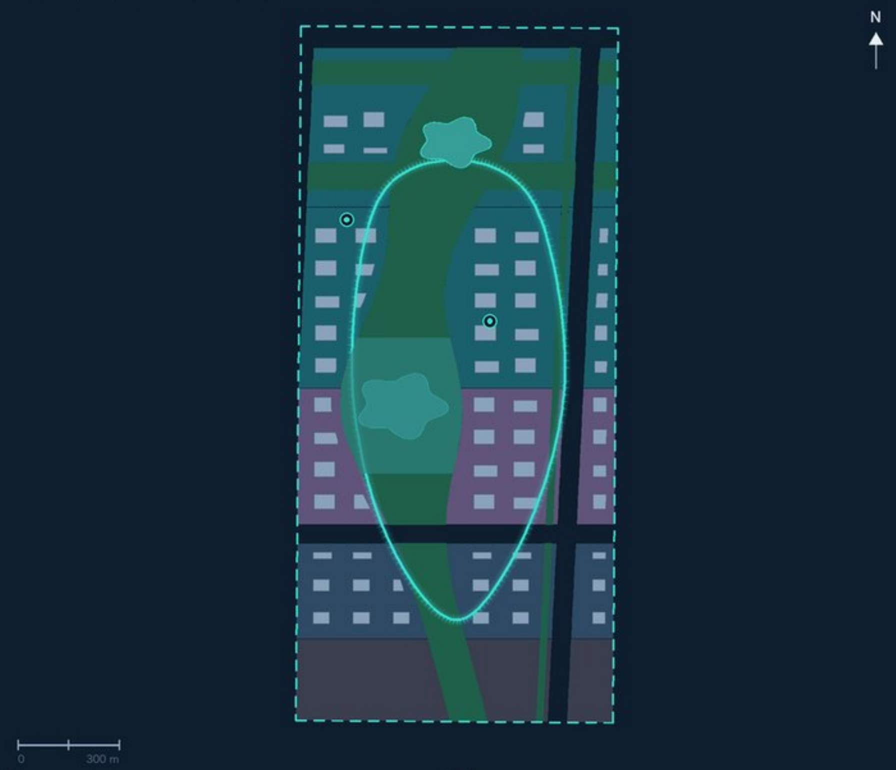
One four-layer protocol landing differently in each key area: R&D base / testbed / showcase window, each closing into its own loop

FIG 03 / KEY AREAS

PROVISIONAL · not an official redline

Zhongzhiyuan AI Innovation Acceleration Area

Protocol R&D and compute base | sense (hook) + knowledge (def)



Key area (provisional)
192.9 ha
announced about 192.1 ha, deviation +0.43%

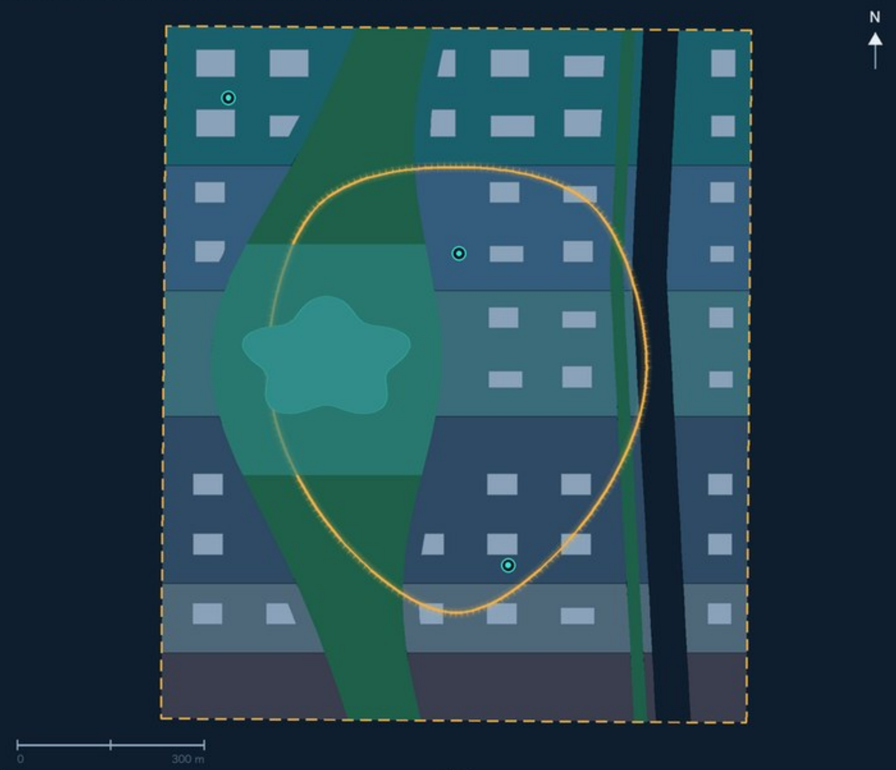
Protocol loop length / conceptual massing
3.24 km
The loop threads 63 conceptual massing footprints (no height or intensity conclusions)

How the four layers land here

- **Sense layer (hook) 17.7 ha** — Compute and sensing facility R&D
- **Knowledge layer (def) 32.6 ha** — Model and knowledge-base R&D
- **Process layer (recipe) 23.2 ha** — Protocol process and industry services
- **Human adjudication 39.4 ha** — R&D talent housing · protocol reserve unit

Beijing AI Origin Community

Protocol testbed (with real residents) | human adjudication + process (recipe)



Key area (provisional)
104.3 ha
announced about 104.3 ha, deviation +0.02%

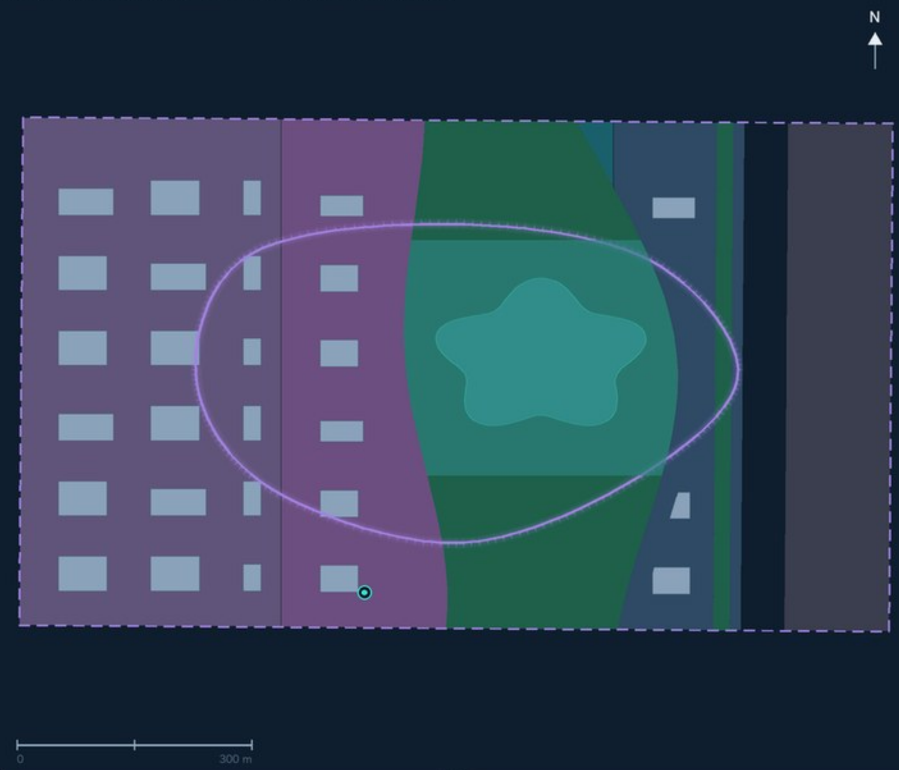
Protocol loop length / conceptual massing
2.08 km
The loop threads 41 conceptual massing footprints (no height or intensity conclusions)

How the four layers land here

- **Sense layer (hook) 15.7 ha** — Scenario sensing and pilot R&D
- **Knowledge layer (def) 12.0 ha** — Education and developer training
- **Process layer (recipe) 10.1 ha** — Community service and protocol deliberation facilities
- **Human adjudication 32.3 ha** — Mixed resident and talent housing · community health and elder-care services · resident co-decision reserve unit

Dazhongsi AI Industry Cluster

Protocol commercialisation and showcase | process (recipe) + knowledge (def)



Key area (provisional)
72.0 ha
announced about 72.0 ha, deviation +0.06%

Protocol loop length / conceptual massing
1.76 km
The loop threads 27 conceptual massing footprints (no height or intensity conclusions)

How the four layers land here

- **Process layer (recipe) 21.6 ha** — Protocol commercialisation and trading services
- **Knowledge layer (def) 11.6 ha** — Cultural display and protocol museum
- **Sense layer (hook) 0.2 ha** — Industrial validation and pilot-scale R&D
- **Human adjudication 15.3 ha** — Station-integrated housing · forecourt protocol reserve unit

Comparison across the three areas

One protocol, different roles: who proposes, who pilots, who showcases

Key area	Protocol role	Layers carried	Dominant land use	AI scenario anchors	Human review gate
Zhongzhiyuan AI Innovation Acceleration Area	Protocol R&D and compute base	Sense (hook) + knowledge (def)	Research-led, with R&D talent housing and reserve	Low-speed delivery and inspection · energy and carbon ledger · open-data access point	Expert and authority review before each protocol release
Beijing AI Origin Community	Protocol testbed (with real residents)	Human adjudication + process (recipe)	Housing + community services + education + health, with resident co-decision reserve	Rail connection and slow-mobility gaps · health service navigation · continuous accessible routes	Resident deliberation with a community-representative veto
Dazhongsi AI Industry Cluster	Protocol commercialisation and showcase window	Process (recipe) + knowledge (def)	Commercial services + cultural display + pilot-scale R&D	Cultural guide · civic proposal and review desk · enterprise service copilot	Compliance review for trading and operations; fact-checking of displayed content

Legend

- Conceptual massing footprint (no height or intensity)
- Protocol node plaza
- Protocol living room (heritage park segment)
- Green and open space
- Protocol loop (closed slow-mobility ring)
- AI scenario pilot unit

Sources: Haidian Branch, Beijing Municipal Commission of Planning and Natural Resources announcement (DATA-SRC-OFFICIAL-ANNOUNCEMENT20260509) + repository provisional boundaries provisional_boundaries.geojson (DATA-SRC-PROVISIONAL-BOUNDARIES20260605); spatial features and metrics recomputed from this proposal's geometry/*.geojson in EPSG:4548.
The rectangular key-area boundaries are inherited from provisional polygons inferred from the official announcement. Their edges must not be read as parcel redlines, road redlines or tenure boundaries. Internal zoning is a conceptual suggestion.
All AI scenarios are labelled "pending human review", "conceptual illustration", "not approved for operation". The authoritative scenario cards are in proposal.md.

Mobility & Blue-Green

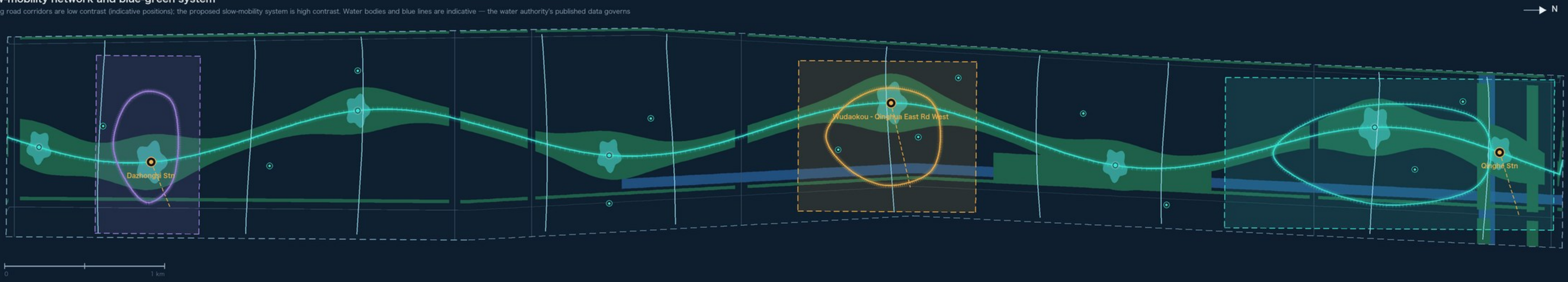
North - south continuity (one slow-mobility bus) × east - west stitching (ten connectors): severance handled in the language of loops and interfaces, with no bridge, tunnel or alignment conclusions

FIG 04 / MOBILITY & BLUE-GREEN

PROVISIONAL · not an official redline

Slow-mobility network and blue-green system

Existing road corridors are low contrast (indicative positions); the proposed slow-mobility system is high contrast. Water bodies and blue lines are indicative — the water authority's published data governs



Continuity in section: accessibility, slow mobility and blue-green all run unbroken

Widths and level changes are conceptual and imply no section design, setback or engineering feasibility conclusion

Continuous pedestrian and accessible frontage

Open ground-floor frontage runs continuously along the loop; ramps and tactile paving are not broken by site entrances

Cycling and low-speed shuttle

The loop doubles as a pilot route for low-speed delivery and inspection robots, with a human takeover gate

Blue-green continuity

Green spine + waterfront scenario belt + buffer greens form one unbroken ecological corridor

Protocol node access

The eight node plazas are shared access points for mobility, scenarios, deliberation and review

Mobility and blue-green metrics

All computed from the geometry, matching metrics.json

Proposed slow-mobility network length
29.58 km
Includes the spine bus, three protocol loops, ten east - west connectors and three rail connectors

Existing road corridor length (indicative)
36.75 km
Positioned roughly from public road names; not a road redline or alignment

Green ratio (self-computed)
19.3 %
220.7 ha of green and open space; not a statutory green-ratio control value

Public space ratio
6.2 %
70.5 ha of public space, measured as a union so no area is double counted

Road land share
12.7 %
144.5 ha of road land (indicative corridor basis)

AI scenario pilot unit
12 sites
6 anchored to the repository's official scenario registry, 6 added by this proposal; all pending human review

East - west stitching · north - south continuity

Interfaces rather than rhetoric: each stitching line is an access point that can be triggered by events and reviewed

CC01	1016 m	Pedestrian / accessible
CC02	1013 m	Pedestrian / accessible
CC03	1009 m	Pedestrian / accessible
CC04	1015 m	Pedestrian / accessible
CC05	1037 m	Pedestrian / accessible
CC06	1190 m	Pedestrian / accessible
CC07	1232 m	Pedestrian / accessible
CC08	1236 m	Pedestrian / accessible
CC09	1222 m	Pedestrian / accessible
CC10	1202 m	Pedestrian / accessible

North - south continuity: a roughly 9.7 km slow-mobility bus links the three areas and two wings into one chain, so an urban event anywhere on it propagates along the whole line. East - west stitching: ten pedestrian and barrier-free connectors cross today's severed interfaces and reattach the spine to the blocks on both sides. Together they form a "bus + interface" urban topology — this proposal's protocol reading of "stitch east - west, connect north - south", with no conclusions on bridges, tunnels, underground space or road alignment.

Legend

- Slow-mobility bus (spine)
- East - west stitching connectors
- Rail station connection (indicative)
- Watercourse blue-line control belt (indicative)
- Green and open space
- Existing road corridor (indicative)

Sources: Haidian Branch, Beijing Municipal Commission of Planning and Natural Resources announcement (DATA-SRC-OFFICIAL-ANNOUNCEMENT20260509) + repository provisional boundaries provisional_boundaries.geojson (DATA-SRC-PROVISIONAL-BOUNDARIES20260605); spatial features and metrics recomputed from this proposal's geometry/*.geojson in EPSG:4548.
Rail station positions are rough indications drawn from clues in the announced brief. They do not represent rail alignments, station positions or engineering conclusions. This proposal gives no conclusions on road alignment, bridges, tunnels or underground feasibility.
Robotics and low-speed shuttle scenarios are all labelled "pending human review", "conceptual illustration", "not approved for operation", with a human takeover gate.

Metrics and evidence chain

Metrics are computed from the geometry, not written; explicit unknowns are never filled with guesses

Metrics & Evidence

Metrics are computed, not written: geometry → metrics.json → drawings / HTML / proposal, and every link can be recomputed independently

FIG 05 / METRICS & EVIDENCE

PROVISIONAL · not an official redline

Core metrics (status = known, all recomputable from the geometry)

Metric	Value	Note
Submitted boundary area	11.413 sq km	announced about 11.4 sq km, deviation +0.11%
Land-use coverage	100.00 %	union(land_use) = site_boundary
Green ratio (self-computed)	19.34 %	220.7 ha
Public space ratio	6.17 %	70.5 ha
Road land share	12.66 %	144.5 ha (indicative basis)
Key areas total	369.3 ha	announced about 368.4 ha, deviation +0.24%

Evidence chain: every link a number passes through

Every link can be verified independently by scripts/spatial_review.py and scripts/visual_review.py

1. **Official announcement**
   Areas, written extents, key-area names and order
   `DATA-SRC-OFFICIAL-ANNOUNCEMENT-20260509`
   ↓
2. **Provisional boundary**
   provisional_boundaries.geojson (official_boundary=false)
   `DATA-SRC-PROVISIONAL-BOUNDARIES-20260605`
   ↓
3. **Proposal geometry**
   geometry/*.geojson · 9 layers · 268 features
   `Generated by this proposal; every feature carries source_type / confidence / geometry_role`
   ↓
4. **Area recomputation**
   Areas and lengths computed after projecting to EPSG:4548
   `area_sqm_declared vs computed geometry, max deviation < 0.001 sqm`
   ↓
5. **Metric registration**
   metrics.json · 51 metrics · known / unknown stated explicitly
   `Unknowns state a reason; no guessed values fill the gaps`
   ↓
6. **Public display**
   Drawings / report / visual/index.html
   `HTML declares data-metric + data-value so machines can compare`

Explicit unknowns: nothing invented, no guessed values

Missing statutory conditions plus this proposal's own discipline together require these metrics to stay empty

total_floor_area_sqm `status = unknown · value = null`
Statutory storey, height and floor-area-ratio controls are absent from the public package. Under this proposal's discipline no development intensity conclusion is given, so total floor area is not inferred.

floor_area_ratio `status = unknown · value = null`
Statutory floor-area-ratio conditions are unreleased. FAR is a statutory intensity control, and this proposal gives no conclusion on it as a matter of discipline.

building_density `status = unknown · value = null`
Statutory building-density conditions are unreleased. The footprints in the geometry cover only the three key areas and are conceptual, so they cannot constitute a site-wide density control value.

building_height_m `status = unknown · value = null`
Official height controls, aviation and landscape view corridors, and heritage height limits are all unreleased. This proposal gives no height conclusion.

Two registers side by side: scenario cards ≠ spatial anchors

Both sides happening to hold 12 entries is a coincidence, not a correspondence; governance objects and spatial carriers are separate registers and must each be checked against their own source file

Governance objects: scenario cards
scenario card table in proposal.md, 12 in total

S-01	Slow-mobility gap diagnosis	PUB-015	S-02	Public space crowding alert	PUB-026
S-03	Open-source release collaboration	no spatial anchor	S-04	Safety evaluation sandbox	no spatial anchor
S-05	Enterprise service copilot	PUB-017	S-06	Elder-friendly and accessible travel	PUB-022
S-07	Cultural guide and narrative	PUB-019	S-08	Data compliance salon	PUB-024
S-09	Low-carbon compute scheduling	PUB-021	S-10	Event safety operations review	PUB-018 / PUB-025
S-11	Low-speed robot delivery pilot	PUB-016	S-12	Health service navigation	PUB-020

Spatial carriers: AI scenario pilot units
geometry/public_space.geojson, 12 in total

PUB-015	Rail connection and slow-mobility gaps	S-01	PUB-016	Low-speed delivery and inspection	S-11
PUB-017	Enterprise service copilot	S-05	PUB-018	Public safety operations review	S-10
PUB-019	Jing-Zhang heritage cultural guide	S-07	PUB-020	Community health service navigation	S-12
PUB-021	Campus energy and carbon ledger	S-09	PUB-022	Continuous accessible route	S-06
PUB-023	Civic proposal and review desk	no matching card	PUB-024	Open data and knowledge-domain access point	S-08
PUB-025	Annual event operations	S-10	PUB-026	Street environment sensing	S-02

What the relationship actually is: 10 cards have a spatial anchor; 2 of them (open-source release collaboration / safety evaluation sandbox) run on existing buildings and institutions and take no anchor of their own. 11 anchors map to a card; 1 (the civic proposal and review desk) anchors the propose - review - merge - rollback governance loop rather than any single card. One card, event safety operations review, lands on two anchors at once. The equal counts are therefore a coincidence, and the metric system keeps the two as separate records.

Geometry self-check

Recomputed by an independent script reading only the written GeoJSON, never build-time objects in memory

●	GeoJSON root type and required properties	268 / 268 passed
●	Geometry validity in EPSG:4548	0 invalid geometries
●	land_use fully tiles the boundary	gap 0.47 sqm (threshold 11.5)
●	land_use has no overlaps	largest pairwise intersection 0.02 sqm (threshold 1.0)
●	Adjacent polygons share vertices	80 adjacent pairs share vertices
●	key_areas disjoint and inside the boundary	overlap 0.00 sqm / 0 outside
●	phasing tiles without overlap	overlap 0.01 sqm / uncovered 0.01 sqm
●	area_sqm_declared matches computed area	max deviation 0.0005 sqm
●	metrics.json matches the geometry	7 / 7 match
○	Official redline	Not yet released; all boundaries provisional

The full self-check report is in geometry-selfcheck.md. The repository's scripts/spatial_review.py returns PASS with only three provisional notices (KEY_AREA_PROVISIONAL), which stem from the organiser's data gap and do not block content scoring.

Spatial evidence for the six agent tasks

Every task maps to features in the geometry, not just to prose

agent.1 Overall concept and coordination
`site_boundary / key_areas / land_use / phasing`
Four-layer protocol section + spine + three loops + eight nodes

agent.2 Full-stack innovation system and ecosystem
`land_use (0802 research · 05 services)`
Zhongguancun service wing interface belt + Zhongzhiyuan R&D base

agent.3 Scenario enablement and a lively city
`public_space (AI_SERVICE_ZONE ×12)`
6 anchored to the official registry + 6 added by this proposal

agent.4 Public space and landmarks
`public_space (living rooms 3 · node plazas 8)`
The heritage park as urban living room; nodes as protocol milestones

agent.5 Cultural fusion narrative
`constraints (heritage belt · advisory) + land_use 0803`
Rail heritage retention belt + cultural display land

agent.6 Event programme and long-term operation
`phasing (release / pilot / extension)`
Protocol version management: release → pilot → extend → revise

Sources: Haidian Branch, Beijing Municipal Commission of Planning and Natural Resources announcement (DATA-SRC-OFFICIAL-ANNOUNCEMENT20260509) + repository provisional boundaries provisional_boundaries.geojson (DATA-SRC-PROVISIONAL-BOUNDARIES20260605); spatial features and metrics recomputed from this proposal's geometry/*.geojson in EPSG:4548.
Metric basis: coordinates exchanged in EPSG:4326, areas and lengths recomputed in EPSG:4548. Every known metric is independently recomputable from the geometry; every unknown metric states why it is missing.
This proposal gives no floor area ratio, building height, building density or development intensity conclusions — both because statutory regulatory conditions are unreleased, and because its own discipline forbids definitive conclusions.

Land-use composition and topology self-check

Areas computed from geometry/land_use.geojson in EPSG:4548; shares use the submitted boundary as denominator

Code	Land-use name	Area (ha)	Share	Composition
0701	Urban housing	217.33	19.04%	
16	Reserved land	186.31	16.32%	
1401	Park green	179.59	15.74%	
0802	Research	159.15	13.95%	
1207	Urban road land	144.48	12.66%	
05	Commercial and business services	128.89	11.29%	
1402	Buffer green	41.14	3.60%	
0804	Education	33.65	2.95%	
1403	Plaza	20.33	1.78%	
0803	Culture	11.64	1.02%	
0702	Urban community service facilities	10.15	0.89%	
0806	Health facilities	8.00	0.70%	
0805	Sports	0.63	0.06%	
	Total	1,141.28	100.00%	

Topology self-check (recomputed by an independent script)

- Total features / required properties complete — 268 / 268
- Invalid geometries in EPSG:4548 — 0
- gap between union(land_use) and the boundary — 0.47 sqm
- union(land_use) outside the boundary — 2.96 sqm
- Largest pairwise land-use overlap — 0.02 sqm
- Adjacent pairs sharing vertices — 80 pairs
- Largest declared-area deviation — 0.0005 sqm
- metrics match the geometry — 7 / 7
- Threshold used here — 11.5 sqm
- Repository spatial_review threshold — 1141 sqm

Reserved land is 16.3%, a deliberate protocol reserve: use and intensity are not fixed in this round but decided jointly by citizens and professional teams under the protocol — the direct land-use counterpart of humans keeping the final decision.

Structural logic: why this is not conventional colour-block zoning

Along: one protocol bus

The Jing-Zhang heritage alignment is abstracted into a continuous green spine with about 29.6 km of unbroken slow mobility, threading the three areas and two wings onto one bus so that an event on any segment travels to the rest.

Across: four-layer protocol section

From west to east: buffer green, knowledge-layer fabric, protocol spine, human-adjudication-layer fabric, process service belt, sense-layer road corridor. The layering is not formal decoration — it places different governance functions where they can see one another.

Local: three closed loops

Each key area closes into a slow-mobility loop with the four layers arranged along it, so that "propose — pilot — evaluate — decide — capture" can run repeatedly at walking scale.

Reserve: clauses not yet decided

186 ha of reserved land sits across the wings and the fabric, standing for clauses of the governance protocol not yet voted on — so a one-off blueprint does not lock in the future.

Renewal projects and phasing

Phasing is protocol version management — release → pilot → extend → revise. It is not an implementation plan or an investment commitment

Protocol release phase 474 ha · 42%

Publish the protocol text and knowledge domains; deliver the three key-area cores plus the first spine segment, establishing an auditable public loop

Pilot phase 406 ha · 36%

The wings and the blocks flanking the spine join the protocol; scenario pilots and human review gates are tested in practice

Extension phase 261 ha · 23%

Existing fabric extends by protocol version; reserve units are decided jointly by citizens and professional teams

Id	Renewal project (conceptual)	Phase	Geometry basis	Notes
RP-01	Connect the first segment of the Jing-Zhang protocol spine	Protocol release phase	land_use · 1401 spine segments	Connect the heritage green spine first to form a working slow-mobility bus
RP-02	Build the three protocol living rooms	Protocol release phase	public_space · living rooms	The park's widened segments become urban living rooms supporting protocol-based booking and co-governance
RP-03	Eight protocol node plazas	Protocol release phase	public_space · node plazas	A public interface anchor shared by release, review, display and deliberation
RP-04	Zhongzhiyuan R&D and compute base block	Protocol release phase	land_use · 0802 Zhongzhiyuan	Spatial carrier for protocol R&D and compute facilities; conceptual massing
RP-05	AI Origin Community resident co-decision unit	Pilot phase	land_use · 16 Origin Community reserve	Housing, community services, education and health organised into a testbed people can join
RP-06	Zhongguancun service wing interface belt	Pilot phase	land_use · 05 service wing	Service space where government services are packaged as standard interfaces
RP-07	Xiaoyuehe scenario wing live testbed	Pilot phase	land_use · 1401 Xiaoyuehe belt	Waterfront parkland as the public carrier for scenario pilots
RP-08	East - west stitching connectors (10)	Pilot phase	roads · stitching interfaces	Pedestrian and accessible links across today's severed interfaces; no bridge or tunnel conclusion
RP-09	Dazhongsi protocol showcase and trading window	Pilot phase	land_use · 05 / 0803 Dazhongsi	Window block for protocol commercialisation and cultural display
RP-10	Co-decision mechanism for protocol reserve units	Extension phase	land_use · 16 reserved land	186 ha of reserve decided jointly by citizens and professional teams under the protocol

AI scenario registry

Each pilot unit anchors AI scenarios in space; anchors and scenario cards are kept as separate registers and are not in one-to-one correspondence; all pending human review and not approved for operation

Scenario id	Scenario	Observed / output	Spatial anchor	Unit area
ai-traffic-walkability	Rail connection and slow-mobility gaps	Transfer walking time · gap locations · accessibility continuity	AI Origin Community / Wudaokou	0.84 ha
robot-delivery-low-speed	Low-speed delivery and inspection	Low-speed lane occupancy and conflict alerts	Zhongzhiyuan	0.71 ha
enterprise-service-copilot	Enterprise service copilot	Policy matching · compliance alerts · scenario access requests	Zhongguancun service wing	0.76 ha
public-safety-operations-review	Public safety operations review	Advance review of event-day crowds and operational risk	Heritage memory segment	0.67 ha
ai-cultural-guide	Jing-Zhang heritage cultural guide	Layered historical narrative guide along the heritage belt	Heritage memory segment	0.63 ha
ai-health-service-navigation	Community health service navigation	Community health and elder-care routes and queue alerts	AI Origin Community	0.60 ha
design-energy-carbon-ledger	Campus energy and carbon ledger	Public ledger of building and facility energy use	Zhongzhiyuan / Qinghe	0.59 ha
design-barrier-free-path	Continuous accessible route	Continuity inspection of wheelchair and low-vision routes	AI Origin Community	0.55 ha
design-civic-proposal-desk	Civic proposal and review desk	Proposal submission · publication · review · rollback	Dazhongsi / south gateway	0.59 ha
design-open-data-kiosk	Open data and knowledge-domain access point	A physical entry where citizens and developers draw from the same data	Xiaoyuehe scenario wing	0.55 ha
design-event-operations	Annual event operations	Venue and crowd scheduling for the open design week and annual meeting	Xiaoyuehe scenario wing	0.63 ha
design-street-environment-watch	Street environment sensing	Street-level sensing of noise, dust and microclimate	Zhongguancun service wing	0.52 ha

Scenario status statement

All 12 scenario units are labelled "pending human review", "conceptual illustration" and "not approved for operation", with a human adjudication gate in the mechanism: machines advise and execute but never replace decisions. Cyan ids come from the repository's official scenario registry (scenarios/*.json); grey ids are added by this proposal. The full scenario cards — trigger conditions, input data, outputs, human review points and failure fallback — are in proposal.md.

Full metric table

metrics.json holds 51 entries; every known metric is recomputable from the geometry and every unknown states why it is empty

Metric (known)	Value	Unit	Formula
site_area_sqm	11,412,825.386	sqm	polygon_area(union(site_boundary), EPSG:4548)
coordinated_research_area_sqm	43,609,232.443	sqm	polygon_area(constraints[scope_id=coordinated_research_area], EPSG:4548)
key_detailed_design_area_sqm	3,692,893.005	sqm	polygon_area(union(key_areas), EPSG:4548)
key_area_count	3	count	count(key_detailed_design_areas)
key_area_zhongzhiyuan_ai_acceleration_area_area_sqm	1,929,201.877	sqm	polygon_area(key_areas[area_id=zhongzhiyuan_ai_acceleration_area], EPSG:4548)
key_area_beijing_ai_origin_community_area_sqm	1,043,236.909	sqm	polygon_area(key_areas[area_id=beijing_ai_origin_community], EPSG:4548)
key_area_dazhongsi_ai_industry_cluster_area_sqm	720,454.219	sqm	polygon_area(key_areas[area_id=dazhongsi_ai_industry_cluster], EPSG:4548)
land_use_total_area_sqm	11,412,827.883	sqm	polygon_area(union(land_use), EPSG:4548)
land_use_coverage_ratio	1.000000	ratio	area(union(land_use)) / area(site_boundary)
land_use_polygon_count	58	count	count(land_use_features)
land_use_area_0701_sqm	2,173,291.731	sqm	polygon_area(union(land_use[land_use_code=0701]), EPSG:4548)
land_use_area_16_sqm	1,863,091.413	sqm	polygon_area(union(land_use[land_use_code=16]), EPSG:4548)
land_use_area_1401_sqm	1,795,920.652	sqm	polygon_area(union(land_use[land_use_code=1401]), EPSG:4548)
land_use_area_0802_sqm	1,591,542.577	sqm	polygon_area(union(land_use[land_use_code=0802]), EPSG:4548)
land_use_area_1207_sqm	1,444,754.671	sqm	polygon_area(union(land_use[land_use_code=1207]), EPSG:4548)
land_use_area_05_sqm	1,288,879.825	sqm	polygon_area(union(land_use[land_use_code=05]), EPSG:4548)
land_use_area_1402_sqm	411,358.643	sqm	polygon_area(union(land_use[land_use_code=1402]), EPSG:4548)
land_use_area_0804_sqm	336,532.668	sqm	polygon_area(union(land_use[land_use_code=0804]), EPSG:4548)
land_use_area_1403_sqm	203,290.017	sqm	polygon_area(union(land_use[land_use_code=1403]), EPSG:4548)
land_use_area_0803_sqm	116,354.695	sqm	polygon_area(union(land_use[land_use_code=0803]), EPSG:4548)
land_use_area_0702_sqm	101,488.149	sqm	polygon_area(union(land_use[land_use_code=0702]), EPSG:4548)
land_use_area_0806_sqm	80,029.569	sqm	polygon_area(union(land_use[land_use_code=0806]), EPSG:4548)
land_use_area_0805_sqm	6,293.703	sqm	polygon_area(union(land_use[land_use_code=0805]), EPSG:4548)
green_space_area_sqm	2,207,279.294	sqm	polygon_area(union(green_space), EPSG:4548)
green_ratio	0.193403	ratio	green_space_area_sqm / site_area_sqm
public_space_area_sqm	704,695.078	sqm	polygon_area(union(public_space), EPSG:4548)
public_space_ratio	0.061746	ratio	public_space_area_sqm / site_area_sqm
road_area_sqm	1,444,754.671	sqm	polygon_area(union(land_use[land_use_code=1207]), EPSG:4548)
road_area_ratio	0.126590	ratio	road_area_sqm / site_area_sqm
slow_mobility_network_length_m	29,575	m	sum(length(roads[geometry_role=design_proposal]), EPSG:4548)
existing_road_corridor_length_m	36,749	m	sum(length(roads[geometry_role!=design_proposal]), EPSG:4548)
building_footprint_area_sqm	243,097.719	sqm	polygon_area(union(building_footprints), EPSG:4548)
building_footprint_count	131	count	count(building_footprints)
phase_1_area_sqm	4,744,927.606	sqm	polygon_area(phasing[phase=phase_1], EPSG:4548)
phase_2_area_sqm	4,061,916.047	sqm	polygon_area(phasing[phase=phase_2], EPSG:4548)
phase_3_area_sqm	2,605,982.760	sqm	polygon_area(phasing[phase=phase_3], EPSG:4548)
phasing_coverage_ratio	1.000000	ratio	area(union(phasing)) / area(site_boundary)
ai_scenario_unit_count	12	count	count(public_space[layer=AI_SERVICE_ZONE])
protocol_node_count	8	count	count(public_space[public_space_type=protocol node plaza])
constraint_feature_count	9	count	count(constraint_features)
ecosystem_case_count	6	count	count(global_innovation_district_cases_in_proposal)
ai_scenario_card_count	12	count	count(scenario_cards_with_trigger_input_output_review_fallback)
persona_count	6	count	count(user_personas_in_proposal)
test_validation_scenario_count	4	count	count(industry_test_validation_scenarios)
pilgrimage_landmark_count	3	count	count(ai_pilgrimage_landmarks)
renewal_project_count	12	count	count(renewal_project_list_entries)
annual_event_count	3	count	count(annual_event_nodes)

Metric (unknown)	Why it is empty
total_floor_area_sqm	Statutory storey, height and floor-area-ratio controls are absent from the public package. Under this proposal's discipline no development intensity conclusion is given, so total floor area is not inferred.
floor_area_ratio	Statutory floor-area-ratio conditions are unreleased. FAR is a statutory intensity control, and this proposal gives no conclusion on it as a matter of discipline.
building_density	Statutory building-density conditions are unreleased. The footprints in the geometry cover only the three key areas and are conceptual, so they cannot constitute a site-wide density control value.
building_height_m	Official height controls, aviation and landscape view corridors, and heritage height limits are all unreleased. This proposal gives no height conclusion.

Data gaps, assumptions and risks

Each item is an explicit treatment forced by a public-data gap or by this proposal's own discipline, not an implicit inference

No official redline

Gap: no official polygon, CAD or GIS redline with a verifiable coordinate system is publicly available.

Treatment: all boundaries use the repository's provisional polygons, marked provisional_constraint and official_boundary=false. Once an official redline is released, every layer, metric, drawing and HTML page must be recomputed as one package.

Road corridor positions

Gap: the announcement gives only road names as written extents, with no alignment data.

Treatment: road corridors are located roughly by name as indicative bands with geometry_role=existing_condition and confidence=low. They are not road redlines, alignments or cross-sections, and this proposal gives no road-alignment conclusion.

Watercourses and heritage extents

Gap: no statutory data is available for blue lines, heritage protection extents or construction control zones.

Treatment: shown as indicative advisory extents with confidence=low; statutory extents are governed by data published by the water and heritage authorities.

Statutory control indicators

Gap: floor area ratio, building height, building density, green ratio and setback controls are absent from the public package.

Treatment: the corresponding metrics are all status=unknown, value=null with a stated reason, and this proposal also deliberately withholds intensity conclusions.

Existing land-use classification

Gap: there is no authoritative mapped data for existing land-use classification.

Treatment: existing fabric is classified conceptually to express structural relationships, with geometry_role=existing_condition and confidence=low, implying no retain/renovate/demolish, tenure or reclassification conclusion.

Heritage park spatial uncertainty

Gap: an OSM background check shows the mapped heritage park is offset from the provisional overall design area.

Treatment: that result neither proves the provisional boundary wrong nor can it be promoted into an official boundary. It is kept as a flag of spatial uncertainty pending an official polygon.

AI scenario compliance

Gap: no scenario is approved for operation, and generative-AI service management rules apply.

Treatment: all are labelled pending human review, conceptual and not approved for operation. Compliance itself is built into the protocol, with a human review gate.

Sources and statements

Every spatial feature carries source_id and source_type, traceable to the sources below

Source id	Sources	What it provides	source_type
DATA-SRC-OFFICIAL-ANNOUNCEMENT-20260509	Announcement of the Haidian Branch, Beijing Municipal Commission of Planning and Natural Resources	Areas of the three scope levels, written extents, and the names and north-to-south order of the three key areas	official_public
DATA-SRC-PROVISIONAL-BOUNDARIES-20260605	Repository provisional boundaries provisional_boundaries.geojson	Provisional rough polygons for the three scope levels and three key areas (official_boundary=false)	agent_inferred_from_public_data
SRC-2023-MNR-LAND-USE-CLASSIFICATION	Project subset of the national Land and Sea Use Classification Guideline	Permitted land-use code values	official_public
REPO-ENUMS	brief/site-package/enums/*.json	Enumerations for layers, road classes, building types and source types	official_public
REPO-SCENARIOS	scenarios/*.json	IDs of 6 officially registered scenarios, used to anchor pilot units	official_public
SELF-GEOMETRY	This proposal's geometry/*.geojson	The direct basis for all spatial features and metrics	agent_generated_design

Statement of use and standing

1. This booklet is a presentation artefact. The authoritative spatial data is geometry/*.geojson, the authoritative metrics are metrics.json and the authoritative narrative is proposal.md; drawings, HTML and PDFs must never be the sole basis for a boundary, area, planning-control or indicator claim.
2. All boundaries are provisional rough boundaries (provisional_constraint, official_boundary=false) and must not serve as an official redline, an approval basis or a basis for precise area recalculation.
3. All spatial proposals are conceptual suggestions / reference schemes / material for professional teams; no conclusions are given on floor area ratio, building height, development intensity, retain-renovate-demolish, road alignment, rail alignment, bridges, tunnels or underground feasibility.
4. Every protocol mechanism is described as proposed, never as implemented; every AI scenario is labelled pending human review, conceptual and not approved for operation.
5. This proposal contains no invented company lists, investment figures, output values or policy commitments; every cited figure carries a source marker.